

DP024
PHYSICS 2
SEMESTER II
SESSION 2023/2024
2 HOURS



UNIT FIZIK
KOLEJ MATRIKULASI SELANGOR

PRE-PSPM DP024 2023/2024

NAME	
MATRIC NO.	
CLASS	

QUESTION	MARKS SCHEME	MARKS
1	16	
2	12	
3	10	
4	14	
5	14	
6	14	
TOTAL	80	

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $x = A \sin \omega t$ | 18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ |
| 2. $\omega = \frac{2\pi}{T} = 2\pi f$ | 19. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$ |
| 3. $v = A\omega \cos \omega t$ | 20. $\tau = RC$ |
| 4. $a = -A\omega^2 \sin \omega t$ | 21. $Q = Q_0 e^{\frac{-t}{RC}}$ |
| 5. $k = \frac{2\pi}{\lambda}$ | 22. $Q = Q_0 \left(1 - e^{\frac{-t}{RC}}\right)$ |
| 6. $v = f\lambda$ | 23. $B = \frac{\mu_0 I}{2\pi r}$ |
| 7. $y(x, t) = A \sin(\omega t \pm kx)$ | 24. $B = \frac{\mu_0 NI}{L}$ |
| 8. $v = \frac{dy}{dt}$ | 25. $B = \frac{\mu_0 I}{2r}$ |
| 9. $y = 2A \cos kx \sin \omega t$ | 26. $\phi = BA \cos \theta$ |
| 10. $v = \sqrt{\frac{T}{\mu}}$ | 27. $\Phi = N\phi$ |
| 11. $I = \frac{dQ}{dt}$ | 28. $\mathcal{E} = -\frac{d\phi}{dt}$ |
| 12. $Q = ne$ | 29. $\mathcal{E} = Blv \sin \theta$ |
| 13. $\rho = \frac{RA}{l}$ | 30. $\mathcal{E} = -NA \frac{dB}{dt}$ |
| 14. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$ | 31. $\mathcal{E} = -NB \frac{dA}{dt}$ |
| 15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | 32. $R = 2f$ |
| 16. $V = IR$ | |
| 17. $C = \frac{Q}{V}$ | |

$$33. \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$34. \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$35. \quad y_m = \frac{m\lambda D}{d}$$

$$36. \quad y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$$

$$37. \quad \Delta y = \frac{\lambda D}{d}$$

Answer all questions.

1. (a) The equation for a mechanical wave is given as $y = 15 \sin (120\pi t - 5\pi x)$; where x and y are in cm and t is in s. Determine the

- (i) frequency
- (ii) wavelength
- (iii) wave speed
- (iv) maximum velocity of the wave
- (v) calculate the particle vibrational velocity at $t = 0.1$ s and $x = 5$ cm.

[10 marks]

- (b) Two waves of the same frequency, amplitude and speed but traveling in opposite directions are represented by the equations,

$$y_1 = 25 \sin (150\pi t - 10\pi x)$$

$$y_2 = 25 \sin (150\pi t + 10\pi x)$$

where x and y are in centimeters and t in seconds.

- (i) write an equation to represent the resultant wave of the superposition of these two waves.
- (ii) calculate the frequency and wave number.

[4 marks]

- (c) A guitar wire has a mass per unit length of μ is $5.16 \times 10^{-3} \text{ kg m}^{-1}$. Under what tension must the wire be kept producing waves with a wave speed of 434 m s^{-1} ?

[2 marks]

2. (a) A wire of cross-sectional 0.9 mm^2 is connected to a voltage, V and the current, I in the wire is measured. At 29°C , the resistance of the wire is 0.45Ω and the resistivity is $1.68 \times 10^{-8} \Omega\text{m}$. If the current measured is 0.5 A , calculate
- The number of electrons every minute.
 - The length of the wire.
 - The voltage in the wire.

[6 marks]

(b)

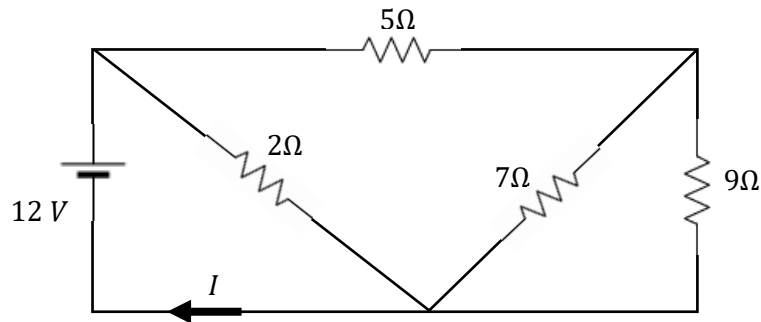
**FIGURE 2.1**

FIGURE 2.1 shows a combination of four resistors connected to 12 V battery. Calculate

- Effective resistance of the resistors.
- The current, I

[6 marks]

3. (a)

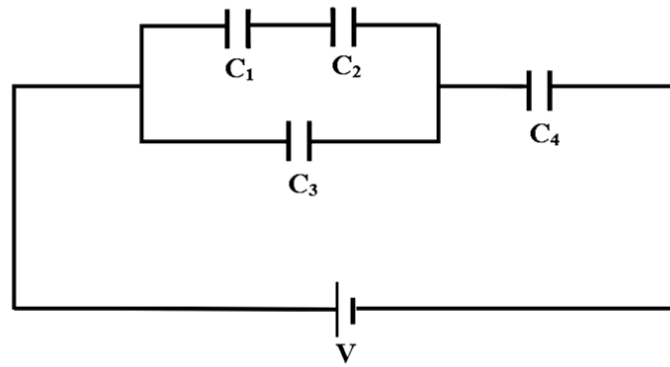
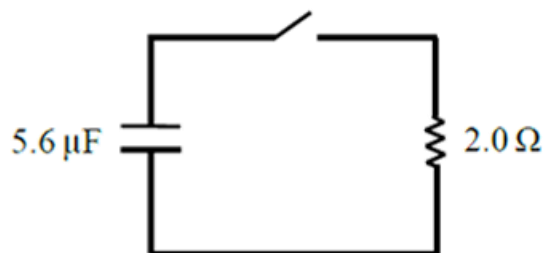
**FIGURE 3.1**

FIGURE 3.1 shows an arrangement of four capacitors connected to a 12 V battery. Given $C_1 = 3 \mu\text{F}$, $C_2 = 6 \mu\text{F}$, $C_3 = 4 \mu\text{F}$, and $C_4 = 6 \mu\text{F}$.

- (i) Calculate the effective capacitance of the circuit.
- (ii) Calculate the charge stored in C_4 .

[6 marks]

(b)

**FIGURE 3.2**

A fully charged $5.6 \mu\text{F}$ capacitor is connected to a 2.0Ω resistor and a switch as in **FIGURE 3.2**. The potential difference across the capacitor is 12.0 V.

- (i) Calculate the time constant of the circuit.
- (ii) Calculate the time for the charge in the capacitor to reduce to one fourth of its initial value after the switch is closed.

[4 marks]

4. (a)

**FIGURE 4.1**

Two long straight wires each carrying a current of 5 A are separated by a distance of 8.0 cm as shown in the **FIGURE 4.1** above. A point P is at a distance of 3.0 cm from wire X . At point P

- (i) calculate magnitude of the magnetic field caused by wire X .
- (ii) calculate magnitude of the magnetic field caused by wire Y .
- (iii) determine the direction of the magnetic field caused by wire X and wire Y .

[7 marks]

(b)

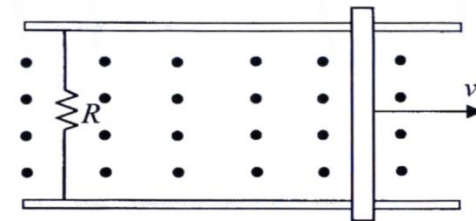
**FIGURE 4.2**

FIGURE 4.2 shows a bar moving on rails to the right with a velocity v in a uniform magnetic field directed out of the page. A resistor R connects the rails.

- (i) What is the direction of induced current in resistor R ?
Explain your answer.
- (ii) State the way to increase the induced current with R fixed.
- (iii) A coil of 100 turns and area 0.5 cm^2 is placed in a changing magnetic field. The rate of change of magnetic fields is 1.08 T s^{-1} . Calculate the induced emf in the coil.
- (iv) A coil of N turns with an area $6.8 \times 10^{-2} \text{ m}^2$ is rotating at frequency 90 Hz in a uniform magnetic field 0.28 T. If the maximum induced emf in the coil is 128.5 V, calculate the value of N .

[7 marks]

5. (a) An object is 20 cm from a convex lens of focal length 12 cm. Calculate the image distance and the magnification of the image formed by the lens. [4 marks]
- (b) An object is placed in front of a concave mirror with 25 cm radius of curvature. A real image twice the size of the object is formed.
(i) Sketch a ray diagram to illustrate the formation of the image.
(ii) Determine the object distance from the mirror. [4 marks]
- (c) An object 4.0 cm is placed 15.0 cm from a convex mirror of a local length 5.0 cm. Determine
(i) the position of the image from the mirror
(ii) the magnification of the image
(iii) the height of the image [6 marks]

6. (a)

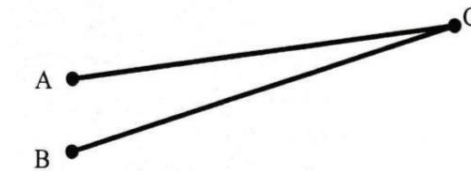


FIGURE 6.1

FIGURE 6.1 shows two paths of in phase coherent lights from points A and B that produce an interference pattern at point C. Determine whether it is a constructive or destructive interference if AC and BC are 6.5λ and 9λ .

[2 marks]

- (b) Two narrow slits separated by 0.03 cm are illuminated by a green light. The screen is placed 1.60 m from the slits and the distance of the first bright fringe from the central bright is 0.3 cm.
(i) Calculate the wavelength of the green light.
(ii) Calculate the distance of the third dark fringes from the central maximum.
(iii) State the changes to the interference pattern if the screen is moved farther from the slits. [6 marks]
- (c) In a Young's double-slits experiment, distance between the screen and the slits is 2.0 m and the wavelength of the light used is 5.0×10^{-7} m. If the distance between the central interference pattern and the tenth bright fringe is 3.4 cm, calculate:
(i) the separation distance between the two slits.
(ii) distance between two adjacent dark fringe fringe
(iii) the separation between the fifth dark fringe and the tenth bright fringe. [6 marks]